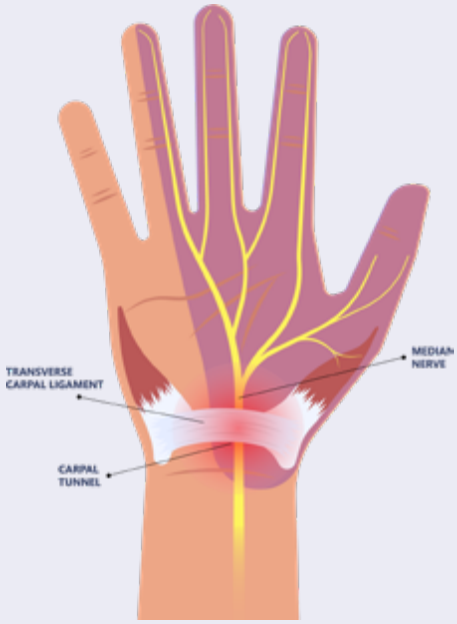
Carpal Tunnel Syndrome

REHAB
REFIT
RETURN TO DUTY

Background

- Carpal Tunnel Syndrome (CTS) is the most common upper extremity nerve compression syndrome.
  It occurs in about 8% of the population.
- CTS happens when the median nerve becomes pinched or compressed in the carpal tunnel, the narrow passage that connects your wrist to the heel of your palm.
- Symptoms include pain, numbness, weakness on the palm side of your thumb, and index and middle fingers. (Figure 1)
- Diagnosis is made through a review of your history, physical exam, and ultrasound evaluation. Additional testing may include x-rays, electromyography (EMG), nerve conduction studies, and MRI.
- Risk factors include diabetes, hypothyroidism, advanced age, obesity, pregnancy, rheumatoid arthritis, and repetitive, forceful wrist work. In addition, biological females are at greater risk than biological males.
- Primary treatments include splinting your wrists at night, stretches, and injections. Surgery is safe and effective, but it's usually considered only after other treatments are unsuccessful.

FIGURE 1



Latest Research

- Computer users do not have an increased risk of CTS when compared to the general population or industrial workers.
- In mild cases, using wrist splints at night, along with forearm and finger stretching, might help to reduce symptoms. (Figure 2)
- For some people, nonsurgical management (splinting, stretching, injections) will fix CTS. However, over half of patients undergoing nonsurgical management end up having surgery within 1 year.

FIGURE 2





Practical Tips for Success

- You might be provided with a wrist brace or instructed to purchase one. Wear it at night to keep your wrist in a neutral position (Figure 2).
- Avoid activities that make your symptoms worse.
- Consider temporary options to avoid activities that make your symptoms worse. For example, instead of using a computer mouse, use the keyboard to navigate (search online for how to do this with your particular operating system), or try using your other hand with your mouse
- You might be referred to occupational or physical therapy for additional rehabilitation.

Rehab Instructions

Carpal Tunnel Stretch



1. Place your palm flat on a wall with your wrist bent to 90 degrees.
2. Gently pull the base of your thumb off the wall.
3. Hold for 30–60 seconds.
4. Complete 3 sets on each wrist 4x a day.

Lumbrical Stretch



1. **Part 1:** Place your palm flat on a table with your wrist bent to 90 degrees.
   a. Pull your fingers up, keeping the underside of your knuckles flat on the table. You should feel this stretch in your wrist.
   b. Hold for 30–60 seconds.

Carpal Tunnel Stretch



2. **Part 2:** Bend your wrist back 90 degrees and curl your fingers in like a claw.
   a. Push your fingertip in towards your back knuckle and hold it for 30 seconds. You should feel this stretch down into your palm.
   b. Repeat on each of your fingers, but not your thumb.
4. Complete 3 sets on each hand 4x a day.

You can also massage the palm of the affected hand with your other thumb to help reduce tension.

Carpal tunnel syndrome: A summary of clinical practice guideline recommendations—using the evidence to guide physical therapist practice. (2019). *Journal of Orthopaedic & Sports Physical Therapy*, 49(5), 359–360. doi:10.2519/jospt.2019.0501

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